

Question Paper

Exam Date & Time: 02-Jan-2026 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 2025 / JAN 2026
SUBJECT: MAT 1171/ MAT_1171 ENGINEERING MATHEMATICS -I

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Using finite difference formula, find the value of $y(1991)$ and $y(2005)$ from the following table. (4)

x	1990	1993	1996	1999	2002
y	110	90	101	98	89

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 1B) Solve $(x^3y^2 + x)dy + (x^2y^3 - y)dx = 0$ (3)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 1C) Solve $\frac{dy}{dx} = \frac{y}{2y \log y + y - x}$ (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2A) Use Runge-Kutta method of order 4 to find the value of $y(0.2)$ given that $\frac{dy}{dx} = 3x + \frac{y}{2}$, with $y(0) = 1$. Take $h = 0.2$ and correct to four decimal places. (4)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 2B) Solve $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = x \cos x$ (3)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 2C) Use method of variation of parameters to solve $\frac{d^2y}{dx^2} + 4y = \tan 2x$ (3)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 3A) Test for consistency and solve by Gauss elimination method (4)
 $3x + 4y - z - 6w = 0$
 $2x + 3y + 2z - 3w = 0$
 $2x + y - 14z - 9w = 0$

$$x + 3y + 13z + 3w = 0$$

Course Outcome: "CO4" / Cognitive Level: "Application"

- 3B) Use numerical differentiation formula to find the value of $y'(0)$ from the following data and correct to 4 decimal places (3)

x	0	2	4	6	8
y	7	13	43	145	367

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3C) Use Simpson's $\frac{3}{8}$ rule to find the area under the curve $y = \sqrt{x^2 + x + 1}$ between the ordinates $x = 1$ and $x = 2.2$. (3)

Take $h = 0.2$ and correct to 4 decimal places.

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 4A) Use Gram Schmidt orthogonalization process, to construct an orthonormal basis from the set of linearly independent vectors $\{a_1 = (1,1,1), a_2 = (1, -2, 1), a_3 = (1, 2, 3)\}$ in R^3 . (4)

Course Outcome: "CO5" / Cognitive Level: "Application"

- 4B) Using Modified Euler's method, find an approximate value of $y(0.4)$, given $\frac{dy}{dx} = x + y; y(0) = 1$. (3)
- Take $h = 0.2$ and carryout 2 iterations for each estimation. Correct to 4 decimal places.

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4C) Use Regula-Falsi method to find a real root of the equation $\sqrt[3]{x} = 10$ between $x = 2$ and $x = 2.5$. Carryout 3 iterations and correct to 3 decimal places. (3)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 5A) Prove that any subset S of a linearly independent set U of a vector space V is linearly independent. Check whether or not, the set of vectors $\{(1,1,0), (1,3,2), (4,9,5)\}$ from R^3 is linearly independent. (4)

Course Outcome: "CO5" / Cognitive Level: "Application"

- 5B) Use Gauss-Seidel method to solve the system of equations (3)
- $$\begin{aligned} x + 17y - 2z &= 48 \\ 30x - 2y + 3z &= 75 \\ 2x + 2y + 18z &= 30 \end{aligned}$$
- Carryout 3 iterations and correct to 4 decimal places.

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5C) Using Rayleigh power method, find the largest eigenvalue and corresponding eigenvector of matrix $A = \begin{bmatrix} 15 & -4 & -3 \\ -10 & 12 & -6 \\ -20 & 4 & -2 \end{bmatrix}$. Use initial vector $X^{(0)} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$. Carry out 3 iterations correct to 3 decimal places. (3)

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Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	MAT_1171:ENGINEERING MATHEMATICS - I

#	Course Outcome	Description	Marks
1	1	CO 1	12
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	8
Total			50

* **NA** - CO Not Selected

Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	MAT_1171:ENGINEERING MATHEMATICS - I

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	8	28(56%)
4	Analysis	7	22(44%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)